

Fundamental Theorem of Calculus

Last week we saw two concepts under the name "integration"

① The anti-derivative.

Given a function $f(x)$ an anti-derivative / indefinite integral is a function $F(x)$ satisfying

$$F'(x) = f(x).$$

and we wrote $\int f(x) dx$ for the (set of) antiderivatives of f .

$$\int f(x) dx = F(x) + C,$$

where C is an arbitrary real constant.

② The definite integral, meaning the ^{signed} area between a function's graph and x -axis.

Formally we defined these, and evaluated some initial examples, using Riemann integration.

On the surface ① and ② seem unrelated. However they are intimately connected by Fundamental Theorem of Calculus.

1. Consider a function f and the definite integral. (a some fixed point)

$$F(x) = \int_a^x f(t) dt, \quad \left(\begin{array}{l} \text{defined via} \\ \text{Riemann} \\ \text{Integration} \end{array} \right)$$

then this $F(x)$ is an anti-derivative for $f(x)$. i.e. $F'(x) = f(x)$.

2. Let $f(x)$ be a function with anti-derivative $F(x)$ (ie $F'(x) = f(x)$) then the definite integral of f can be evaluated by

$$\int_a^b f(x) dx = F(b) - F(a)$$

This is the version we typically use to evaluate definite integrals in practice.

An argument to prove part 2.

Consider a function f and its anti-derivative $F(x)$, ie. $F'(x) = f(x)$.

Consider the "linear approximation with associated error" for $F(x)$.

For any point c . linear approx.

$$F(c+h) = F(c) + hF'(c) + h\varepsilon(h)$$

where $\varepsilon(h) \rightarrow 0$ as $h \rightarrow 0$

Start with the Riemann integral $\int_a^b f(x) dx$. Use equally spaced partition $x_{\bar{i}}, \bar{i}=0, \dots, n$ defined by $x_{\bar{i}} = a + \bar{i} \frac{(b-a)}{n}$, so $x_0 = a, x_n = b$. the width of each rectangle is $\frac{b-a}{n}$ and use left-hand-rule.

$$x_{\bar{i}}^* = a + (\bar{i}-1) \frac{(b-a)}{n} = x_{\bar{i}-1}$$

$$\int_a^b f(x) dx = \int_a^b F'(x) dx = \lim_{n \rightarrow \infty} \left(\sum_{\bar{i}=1}^n F'(\underbrace{a + (\bar{i}-1) \frac{(b-a)}{n}}_{x_{\bar{i}-1}}) \underbrace{\frac{(b-a)}{n}}_h \right)$$

$$= \lim_{n \rightarrow \infty} \left(\sum_{\bar{i}=1}^n \left(F(x+h) - F(x) - h \varepsilon(h) \right) \right)$$

using lin. approx.
with error result.

$$= \lim_{n \rightarrow \infty} \left(\sum_{\bar{i}=1}^n F(x_{\bar{i}}) - F(x_{\bar{i}-1}) - h \varepsilon(h) \right)$$

Ch 6. Techniques of anti-differentiation.

Thanks to the FTC. we can use our knowledge of differentiation. to give us techniques for evaluating definite integrals, by first finding appropriate anti-derivative.

such properties:

linearity, product rule, chain rule, ---

Σ
↓
int. by
parts.

Σ
↓
int. by
substitution.

6-1 Substitution Comes from the chain rule.

For example. Consider a function.

$$f(x) = (x^2 + 3x - 5)^{10}$$

$$= h(g(x)), \text{ where}$$

$$g(x) = x^2 + 3x - 5$$

$$h(x) = x^{10}$$

The chain rule tells us

$$h'(x) = 10x^9$$

$$g'(x) = 2x + 3$$

$$f'(x) = h'(g(x)) g'(x)$$

$$= 10(x^2 + 3x - 5)^9 (2x + 3)$$

$$= (20x + 30)(x^2 + 3x - 5)^9$$

So if we're confronted with an integral.

$$I = \int (20x + 30) (x^2 + 3x - 5)^9 dx.$$

Can use a substitution

$$u = x^2 + 3x - 5$$

$$\Rightarrow du = 2x dx + 3 dx = (2x + 3) dx$$

$$\Rightarrow dx = \frac{du}{(2x + 3)}$$

So the integral I will become.

$$I = \int (20x + 30) u^9 \frac{du}{(2x + 3)}$$

$$= \int 10 u^9 du$$

$$= u^{10} + C$$

$$= (x^2 + 3x - 5)^{10} + C.$$

where C is the
arbitrary constant
relating anti-derivatives

Another example

$$I = \int x \sin(x^2 + 5) dx$$

here we could use.

According to technique laid in Key Idea 6.5.2, $r(x)$ will have an alternative expression as

$$r(x) = \frac{A}{x-1} + \frac{B}{(x+2)} + \frac{C}{(x+2)^2}$$

for some suitable values of $A, B, C \in \mathbb{R}$, which can be found as follows,

$$= \frac{A(x+2)^2 + B(x-1)(x+2) + C(x-1)}{(x-1)(x+2)^2}$$

$$= \frac{(A+B)x^2 + (4A+B+C)x + (4A-2B-C)}{(x-1)(x+2)^2}$$

Comparing this numerator to the original one, 1.

$$\Rightarrow \begin{cases} A+B=0 \\ 4A+B+C=0 \\ 4A-2B-C=1. \end{cases}$$

Solve this system of equations to find the solution A, B, C . We have the

Gaussian Elimination / row reduction algorithm for solving such systems.

Augmented matrix

$$\begin{array}{c} A \quad B \quad C \quad \text{rhs} \\ \left(\begin{array}{cccc} 1 & 1 & 0 & 0 \\ 4 & 1 & 1 & 0 \\ 4 & -2 & -1 & 1 \end{array} \right) \end{array}$$

Use elementary row ops to reduce this to row-echelon form. so that the solution can be easily read from the matrix.

$$\left(\begin{array}{cccc} 1 & 1 & 0 & 0 \\ 0 & -3 & 1 & 0 \\ 0 & -6 & -1 & 1 \end{array} \right)$$

$$r_2' = r_2 - 4r_1$$

$$r_3' = r_3 - 4r_1$$

$$\left(\begin{array}{cccc} 1 & 1 & 0 & 0 \\ 0 & 1 & -1/3 & 0 \\ 0 & -6 & -1 & 1 \end{array} \right)$$

$$r_2' = -\frac{1}{3} r_3$$

$$\left(\begin{array}{cccc} 1 & 0 & 1/3 & 0 \\ 0 & 1 & -1/3 & 0 \\ 0 & 0 & -3 & 1 \end{array} \right)$$

$$r_1' = r_1 - r_2$$

$$r_3' = r_3 + 6r_2$$

$$\left(\begin{array}{cccc} 1 & 0 & 1/3 & 0 \\ 0 & 1 & -1/3 & 0 \\ 0 & 0 & 1 & -1/3 \end{array} \right)$$

$$r_3' = -\frac{1}{3} r_3$$

$$r_1' = r_1 - \frac{1}{3} r_3$$

$$r_2' = r_2 + \frac{1}{3} r_3$$

$$\left(\begin{array}{cccc} 1 & 0 & 0 & 1/9 \\ 0 & 1 & 0 & -1/9 \\ 0 & 0 & 1 & -1/3 \end{array} \right)$$

This final matrix represents the equations

$$A = 1/9$$

$$B = -1/9$$

$$C = -1/3$$

So we ~~the~~ now have the partial fraction expansion for $r(x)$ as

$$r(x) = \frac{1/9}{x-1} - \frac{1/9}{x+2} - \frac{1/3}{(x+2)^2}$$

$$I = \int r(x) dx$$

$$= \int \left(\frac{1/9}{x-1} - \frac{1/9}{x+2} - \frac{1/3}{(x+2)^2} \right) dx$$

$$= \frac{1}{9} \int \frac{1}{x-1} dx - \frac{1}{9} \int \frac{1}{x+2} dx - \frac{1}{3} \int \frac{1}{(x+2)^2} dx$$

These three integrals are known standard integrals.

$$= \frac{1}{9} \ln(x-1) - \frac{1}{9} \ln(x+2) - \frac{1}{3} \frac{-1}{x+2}$$

$$= \frac{1}{9} \ln(x-1) - \frac{1}{9} \ln(x+2) + \frac{1}{3(x+2)}$$

